What produces a capturable complexity gap: one representative economy per model and group

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Scope. Every economy with a protocol run (the committed record in `variants/results/economy_table.csv` plus the October 6–15, 2026 Quest campaign in `bop_runs/results`) and the reduced-form screens. The main tables hold one representative per model per group, built so that the same change is made in all three models; everything else that was run is in the appendix (Section 8). The tables are generated by `variants/campaign_tables.py` from the oracle and estimator files. Every campaign economy ran ten seeds (`gvypk3` twenty; the reduced form five per cell).

Reading the tables. Every table has the same columns. $\mathrm{SR}_{\max}$, the *linear ceiling* and the *room* are the population investor's conditional Sharpe ratios over the evaluation window: the linear ceiling is the best Sharpe a model linear in the ranks of the characteristics reaches with the true conditional moments, the room is the best nonlinear ceiling (random features, bins, quadratic) minus that linear ceiling, and *room % of linear* is the room as a percentage of the linear ceiling. That is the theoretical scope for nonlinearity, before any estimation. Sharpe ratios add in quadrature, so the room understates an interaction that sits on a large linear base; *interaction SR* is the nonlinear ceiling's Sharpe orthogonal to the linear one, $\sqrt{\mathrm{nl}^2 - \mathrm{lin}^2}$, the size of the interaction itself, and *DKKM* $\perp$ is DKKM's Sharpe orthogonal to the best classical method, signed like the gap. *Market share* is the equal-weighted market's share of $\mathrm{SR}_{\max}$. *DKKM*, *FMR* (Fama–MacBeth), *FF* (Fama–French factors), *linear* (the best of rank and level regressions, with and without the market) and *CAPM* (the estimated market line) are each method's out-of-sample Sharpe at the 360-month window. All Sharpe ratios are monthly (multiply by $\sqrt{12}$ for annual), so a gap of 0.01 is 0.035 a year. The *fair gap* is DKKM minus the best of the four classical methods, averaged across seeds, with in parentheses its t-statistic (the mean over its standard error across seeds; ten seeds unless stated); "+" counts the seeds in which it is positive. A dagger marks an oracle that predates the evaluation-window ceilings.

1 The answer

Each model has a spine: its published baseline, and one economy in which a priced state y enters firm cash flows through an interaction of two firm variables (group 3b). Every other group is one change to that spine, the same change in all three models. Only the interaction produces a fair gap. With one type, every firm's exposure varies with one firm variable (3a), the premium is monotone in one characteristic's rank, and room is 5 to 10% of the linear ceiling with no gap; a crash-only factor (1) gives room of 3 to 12% and no gap; a quadratic price of the model's own risk (2) gives room of 0 to 14% and no gap; the interaction economies (3b) have room of 16 to 47% and gaps of +0.008 to +0.020 (t 2.7 to 5.2). More priced factors (4) help only where each factor carries its own interaction: in the reduced form, factors that load on products of partly hidden states give room and a gap that rise with

the number of factors and with the noise hiding the states, to +0.29 at ten factors, while factors that load linearly on observed characteristics give nothing at any K. The structural two-state economies (appendix) add no interaction: a second state through the same channel dilutes the exposure factor (KP14's room falls from 24% to 15%), and the designs that raise BGN's and GS21's gaps do so by giving each type its own state's premium, time variation in a sort rather than a second product.

The weight analysis (Section 6) shows the mechanism directly. In the 3b economies the true weight's slope in book-to-market depends on size (KP14, BGN) or on profitability (GS21); DKKM's fitted weight surface bends the same way, its nonlinear part correlating 0.77, 0.84 and 0.49 with the truth's, and in KP14 the interaction it recovers has the truth's size (0.107 against 0.106); Fama–MacBeth's surface is parallel lines by construction; and in the benchmarks there is no bend to find.

Economically: every factor model is firm exposure times aggregate shock, so the interaction that matters is not with the aggregate state but within the exposure. A capturable gap needs a firm's exposure to a priced shock to be the product of two things that different characteristics reveal, "what the firm is exposed to" (its type, or its productivity) times "how levered the firm's claim is to it" (growth options in KP14, the project portfolio in BGN, the coupon in GS21). The size of a structural gap is then bounded: the part of the premium no linear portfolio holds is the dispersion of the exposure factor relative to its mean, and a multiplicative channel keeps that ratio below one. That is why every structural gap is modest; why the GS21 economy in which leverage acts through the equity claim carries an interaction as large as KP14's (interaction SR 0.33) yet a near-zero gap, on a linear base of 0.81 (Section 3b); and why the imposed GS21 product (productivity times coupon, mean zero) and the reduced form's latent products reach gaps of +0.02 to +0.3: they have no additive part for a linear portfolio to hold.

group	the one change to the spine	room % of linear	fair gap
1. distribution of shocks	a crash-only cash-flow factor	3–12	none
2. price of risk	quadratic in the model's own state	0–14	none
3a. single-characteristic exposure	one type exposed to y	5–10	none
3b. interaction	three types (a product in GS21) exposed to y	16–47	+0.008 to +0.020
4. larger factor model (reduced form)	K factors loading linearly on characteristics / on products of latent states	0–1 / 190–660	none / +0.03 to +0.29
structural second priced state (appendix)	a second chain in each model	15–65	+0.005 to +0.056

The baselines every group is read against:

Table 1: Baselines. The three models as published, one type each: the floor every group is read against.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bgnbase` – baseline, no priced state beyond the rate	0.301	54%	0.250	+0.027	11	0.12	0.238	0.213	0.208	0.238	0.162	-0.0001 (t -0.0)	+0.00	6/10
GS21	`gsbase` – baseline	0.293	95%	0.291	+0.001	0	0.03	0.283	0.265	0.261	0.286	0.278	-0.0028 (t -4.7)	-0.03	2/10
KP14	`kpbase` – baseline, one type, no priced state	0.229	35%	0.221	+0.004	2	0.04	0.208	0.193	0.195	0.209	0.078	-0.0010 (t -0.6)	-0.00	6/10

2 A different distribution of shocks: a crash-only factor

A cash-flow factor y (unit variance, half-life two years) multiplies every firm's cash flows as e^{by}, with one exposure b per model: 0.035 in BGN's project cash flows, $0.10\,\tilde{z}$ in GS21's output (the product economy's exposure without the coupon), 0.07 in KP14's project cash flows. Its Gaussian shocks

carry no price. Its only priced event is a crash: each month, with probability 1%, y jumps to -3.5 standard deviations, and the kernel weights that transition by e^2 (about 7% a month risk-neutral); y then recovers at its ordinary mean reversion. Against the single-type economy of group 3a, the only change is that the Gaussian price is replaced by the crash.

Table 2: Group 1. A different distribution of shocks: a crash-only factor. Every firm's cash flows load on a factor y whose only priced shock is a rare crash (1% a month to -3.5 sd, kernel weight e^2).

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bvy1j` – crash-only factor: $e^{0.035y}$, the same crash	0.324	50%	0.278	+0.033	12	0.14	0.265	0.239	0.235	0.262	0.160	+0.0003 (t 0.1)	+0.01	6/10
GS21	`gvyz10j` – crash-only factor on $0.10\,\bar{z}\,y$: y unpriced except the 1%/month jump to -3.5 (weight e^2)	0.386	82%	0.374	+0.011	3	0.09	0.358	0.342	0.334	0.363	0.316	-0.0047 (t -4.0)	-0.05	1/10
KP14	`vy1j` – crash-only factor: $e^{0.07v}$, y unpriced except a 1%/month jump to -3.5 (weight e^2)	0.336	21%	0.277	+0.012	4	0.08	0.259	0.241	0.249	0.265	0.066	-0.0057 (t -2.3)	-0.04	2/10

Verdict. A crash changes the distribution of a shock every firm bears in the same proportion. The cross-section of premia it creates is a sort on the one firm state that scales the exposure, so the linear ceiling takes nearly all of it: room is 12% of the linear ceiling in BGN, 3% in GS21 and 4% in KP14, and the fair gap is zero or negative in all three (+0.000, −0.005, −0.006). Adding the same crash to the interaction economies instead (Appendix A1) leaves their room and gap at the control's in BGN and GS21 and halves KP14's.

3 A different price of risk: quadratic in the model's own state

The price of each model's own aggregate risk is multiplied by $1 + 0.25\,s^2$, capped at 3, where s is a standardized aggregate state: GS21's productivity x, and for KP14, whose own states are random walks, an unpriced stationary state y that no firm is exposed to. The multiplier never falls below 1 because KP14's growth-option discount turns negative once the price of x-risk drops below about 0.82 of its value. In BGN the price of the market shock σ_z is multiplied by the same function of the standardized rate, binned to seven equiprobable levels of the rate (1.00, 1.03, 1.17 and 1.68 from the middle outward) so that project values keep a closed form; the date-k cash flow is priced at the bin of the rate one period before it pays.

Table 3: Group 2. A quadratic price of the model's own risk. The price of the baseline's aggregate risk becomes $1 + 0.25s^2$ times its value, capped at 3, s a standardized aggregate state.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bq25` – $\sigma_z \times \min(1 + 0.25\,s^2, 3)$, s the standardized rate, 7 bins	0.411	36%	0.346	+0.049	15	0.18	0.339	0.303	0.293	0.335	0.149	+0.0034 (t 0.7)	+0.00	4/10
GS21	`gq25` – $\gamma_x \times \min(1 + 0.25\,\bar{x}^2, 3)$	0.345	95%	0.343	+0.002	0	0.03	0.334	0.322	0.313	0.338	0.326	-0.0038 (t -5.1)	-0.05	0/10
KP14	`kpq` – $\gamma_x \times \min(1 + 0.25\,y^2, 3)$, y unpriced	0.260	53%	0.251	+0.003	1	0.04	0.240	0.226	0.225	0.242	0.137	-0.0021 (t -1.5)	-0.02	4/10

Verdict. A price of risk that moves with the aggregate state rotates the premium across states and leaves it linear in the ranks within one. The quadratic price gives room under 1% of the linear ceiling in GS21 and KP14, and 14% in BGN, where the rate is itself a priced state so a price that rises with its distance from the mean rotates the premium with it; the fair gap is zero in all three (+0.003, −0.004, −0.002). The regime prices tried earlier (appendix) do the same, and BGN's fast regime gave the group's one small positive number (+0.008), which a ranks-times-state linear benchmark takes.

4 Different exposures to aggregate risk

4.1 Exposure that a single characteristic captures

The priced state y is added with one type: every firm's cash flows load on it with the same exponent (the group 1 exposures, now with y's Gaussian shocks priced at 2.5). A firm's exposure to y then varies across firms only through one firm state: the mix of assets in place and growth option in BGN,

productivity in GS21 (the product economy's exposure without the coupon), the growth-option share in KP14.

Table 4: Group 3a. Exposure captured by a single characteristic. A priced state y with one type: a firm's exposure varies only through one firm state (project mix, operating leverage, growth-option share).

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bvy1ch` – one type, project cash flows $\times e^{0.035y}$	0.336	54%	0.297	+0.029	10	0.13	0.283	0.252	0.255	0.281	0.181	+0.0017 (t 0.8)	+0.02	7/10
GS21	`gvyz10` – one type, output $\times e^{0.10\,\tilde{z}\,y}$; the product economy without the coupon	0.854	50%	0.800	+0.043	5	0.27	0.793	0.806	0.743	0.809	0.421	-0.0170 (t -5.4)	-0.15	1/10
KP14	`vy1` – one type, cash flows $\times e^{0.07y}$	0.466	12%	0.369	+0.019	5	0.12	0.355	0.335	0.340	0.360	0.047	-0.0050 (t -2.9)	-0.05	1/10

Verdict. A monotone function of one firm state, however nonlinear, is linear in the rank of the characteristic that reveals it. Room is 10% of the linear ceiling in BGN and 5% in GS21 and KP14, and the fair gap is zero or negative in all three (+0.002, −0.017, −0.005); in GS21 the productivity exposure alone carries a large premium (SR_{max} 0.85) that the linear ceiling captures almost entirely. The same holds for every other single-characteristic wiring tried (Appendix A3), including ones with far more room: room is necessary, not sufficient. The absence of room is not an artefact of the rank standardization DKKM applies to its random features after the sines and cosines: recomputing the ceilings on the saved panels with the raw features (two seeds per economy, `variants/mlp/room_raw.py`) gives room of 4% / 6% / 8% of the linear ceiling in `vy1` / `gvyz10` / `bvy1ch` against 5% / 6% / 8% with the standardization, and 19% / 62% / 10% against 26% / 61% / 11% in the 3b economies; the quadratic and bin ceilings, which never standardize, say the same.

4.2 Exposure as an interaction of several characteristics

The same y with heterogeneity in what is exposed to it: three types of firms with exposures 0.01 / 0.035 / 0.07 (BGN) and 0.02 / 0.07 / 0.14 (KP14), level-compensated so that value at $y = 0$ does not reveal the type; and in GS21, where types never produced room (Appendix A3), the product of standardized productivity and the standardized coupon written into the production function with coefficient 0.10. Exposure is then a product: type × project mix (BGN), type × growth-option share (KP14), productivity × leverage (GS21), each factor revealed by a different characteristic.

Table 5: Group 3b. Exposure as an interaction of characteristics. The same y with three types (BGN, KP14) or a product written into production (GS21): exposure = type or productivity × how levered the claim is.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bvych0` – y as a 21-node chain, three types (control)	0.576	37%	0.341	+0.055	16	0.20	0.334	0.285	0.292	0.326	0.216	+0.0076 (t 2.7)	+0.05	8/10
GS21	`gvyp10` – product exposure $\tilde{z}\bar{b}$ at 0.10 (control)	0.650	8%	0.447	+0.194	47	0.46	0.550	0.189	0.333	0.531	0.018	+0.0196 (t 2.8)	+0.12	9/10
KP14	`vyg25` – three types on y, price 2.5 (control)	0.495	11%	0.346	+0.082	24	0.25	0.350	0.292	0.320	0.335	0.044	+0.0148 (t 5.2)	+0.09	10/10

Verdict. Every interaction economy has a positive mean gap with most seeds positive (t of 2.7 to 5.1 across seeds), at room of 16 to 47% of the linear ceiling. The gaps are statistically clear and economically modest: 2 to 4% of the linear Sharpe at these three representatives; the scaled GS21 versions in the appendix reach 11 to 22%. The premium is multiplicatively separable in the ranks, $f(A)\,g(B)$; a linear model in the ranks fits only additive functions, random features span products, and DKKM recovers 15 to 30% of the room as a gap. Going from 3a to 3b changes nothing but the second dimension of heterogeneity, and that is what creates the gap. The GS21 product ladder (Appendix A4) shows the room hump-shaped in the product's size and the gap rising to +0.08. The GS21 product is imposed: the coupon multiplies the exposure inside the production function. Its structural counterpart, where the exposure $0.10\,\tilde{z}\,y$ sits in cash flows and leverage acts only through the equity claim, was tested by fixing each firm's capital structure as a type drawn independently of productivity (`gvyzlf`: coupons 0.16, 0.42 and 0.68 per unit capital, debt-to-equity 0.25 to 3.25,

no refinancing; four variants in the appendix). The interaction is there, the true premium's slope in leverage flips sign across profitability quintiles, but the equity claim amplifies the y premium by only 20 to 40% around a large additive productivity part, so room stays at 6 to 8% of the linear ceiling and the gap at +0.005 (t 1.8). The imposed product has no additive part, which is where its 43% comes from. Three levers on `gvyzlf` fail to change this (appendix). Valuing the debt as riskless (`gvyzlr`) leaves every number unchanged, so the debt was not absorbing y risk; a more persistent y (`gvyzlp`, κ_y 0.10) halves $\mathrm{SR}_{\max}$ and the interaction with it; and cutting the price of the aggregate productivity shock from 0.5 to 0.1 (`gvyzlg`, with its own benchmark `gsbg1` and 3a comparator `gvyz10g`) lowers the market's share from 50% to 27% but leaves the linear ceiling at 0.72, because the base is the priced productivity level itself rather than the market: room 9%, gap −0.006. The interaction's own Sharpe is 0.31 to 0.33 in all of them, the size seen in KP14's and BGN's 3b economies (table column *interaction SR*); what differs is the base it sits on. Two further wirings close the question from the other side: fixing the product's sign with a leverage-level term (`gvyp10c`, exposure $0.10\,\tilde{b}(\tilde{z}+2)$) hands the market a large negative premium and leaves room of 1%, and KP14's own recipe, three latent exposure types times three fixed leverage types (`gvytl`, and `gvytlg` at a fifth of the aggregate shock's price), gives the market 72 to 84% of the attainable Sharpe and room of 1 to 3%. Every exposure with a nonzero cross-sectional mean is carried by GS21's equal-weighted market; the product is the one wiring whose mean is zero, which is why it alone has room.

5 Larger factor models

More priced factors are tested where the structure is controlled: the reduced form below adds K priced factors to a market factor, with the total price of risk held across K, and varies only whether the factors load linearly on the observed characteristics or on products of partly hidden states (setup in Section 7). The structural version, a second priced state in each of the three models, is in the appendix (Table 9).

Table 6: Reduced form: priced factors loading linearly on observed characteristics against products of latent states. Five seeds per cell, the total price of risk fixed across K (per-factor price $\propto 1/\sqrt{K}$); K = number of priced factors; the lfh rows load products of pairs of the five latent processes, observed with noise of sd 0 (n0) or 1 (n1).

configuration	$\mathrm{SR}_{\max}$	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	+
kfh_lin_1 – K=1 factor on observed characteristics, linear loadings	0.598	8	0.598	+0.000	0	0.01	0.590	0.584	0.581	0.595	0.041	-0.0067 (t -1.7)	0/5
kfh_lin_4 – K=4	0.592	8	0.588	+0.003	0	0.06	0.555	0.578	0.567	0.575	0.041	-0.0237 (t -3.1)	0/5
kfh_lin_16 – K=16	0.385	13	0.383	+0.001	0	0.03	0.352	0.357	0.353	0.365	0.041	-0.0128 (t -1.2)	2/5
lfh_L5_K1_n0 – K=1 product of latent states, no observation noise	0.598	8	0.208	+0.389	187	0.56	0.570	0.156	0.539	0.175	0.041	+0.0311 (t 4.0)	5/5
lfh_L5_K2_n0 – K=2	0.598	8	0.144	+0.453	314	0.58	0.564	0.111	0.520	0.115	0.041	+0.0443 (t 6.7)	5/5
lfh_L5_K3_n0 – K=3	0.598	8	0.106	+0.490	464	0.59	0.566	0.059	0.507	0.084	0.041	+0.0589 (t 9.0)	5/5
lfh_L5_K5_n0 – K=5	0.598	8	0.092	+0.502	549	0.59	0.543	0.048	0.423	0.064	0.041	+0.1193 (t 21.6)	5/5
lfh_L5_K10_n0 – K=10	0.598	8	0.089	+0.503	568	0.58	0.507	0.047	0.250	0.067	0.041	+0.2572 (t 39.9)	5/5
lfh_L5_K1_n1 – K=1 product of latent states, observation noise 1.0	0.598	8	0.167	+0.416	249	0.56	0.526	0.096	0.433	0.137	0.041	+0.0930 (t 3.8)	5/5
lfh_L5_K2_n1 – K=2	0.598	8	0.111	+0.469	423	0.57	0.523	0.072	0.420	0.082	0.041	+0.1025 (t 6.8)	5/5
lfh_L5_K3_n1 – K=3	0.598	8	0.091	+0.486	537	0.57	0.528	0.057	0.391	0.073	0.041	+0.1371 (t 10.4)	5/5
lfh_L5_K5_n1 – K=5	0.598	8	0.075	+0.496	664	0.57	0.496	0.038	0.309	0.060	0.041	+0.1866 (t 26.3)	5/5
lfh_L5_K10_n1 – K=10	0.598	8	0.079	+0.479	607	0.55	0.456	0.052	0.171	0.077	0.041	+0.2854 (t 24.8)	5/5

Factors that load linearly on observed characteristics leave no room at any K and a negative gap; factors that load on products of partly hidden states give room and a gap that grow monotonically with K and with the observation noise that hides the states. Bounded and curvature loadings on observed characteristics are in the appendix (the RF rows of Table 8).

6 What the estimated weights look like: the nonlinearity DKKM picks up

The Sharpe tables say DKKM beats the linear methods in the 3b economies; this section asks what its weights do that the linear methods' weights cannot. It applies the neural-network weight analysis (Pruitt's `MLP Analyze` pipeline, `external/MLP Analyze/BOP campaign/run_bop.py`) to the three benchmarks and the three 3b economies. For every economy, the per-firm weights of each method are rebuilt from the saved panels exactly as `run_estimators.py` formed them (`bop_runs/mlp/extract_weights.py`; the rebuild is accepted only if it reproduces the recorded per-month Sharpe ratios, and it does so to 10^{-16}): the true SDF weights (the population investor's conditional mean-variance portfolio), the Fama–MacBeth weights, the Fama–French weights and the DKKM weights ($P = 3600$ random features, the two-draw ensemble, the economy's best κ), over the 125 evaluation months of every seed. A 5×32 ReLU network is then fitted to each method's weight as a function of the five characteristics, rank-standardized within each (seed, month) cross-section; the network is a smoother, and what is read off it is the shape of the weight surface, not its fit. Three statistics summarize each surface against the truth's. The *correlation with truth* is the cross-sectional correlation of the method's weights with the true weights, averaged over cross-sections (first for the weights themselves, then for the fitted surface). The *nonlinear share* is the part of the fitted surface's variance that a linear regression on the five ranks within each cross-section leaves: the part of the surface no portfolio linear in the ranks can hold. The *correlation of nonlinear parts* is the cross-sectional correlation of that remainder with the truth's remainder: whether the method's departures from linearity are the truth's departures or noise. Its t-statistic is across seeds. Two figures per model show the surfaces themselves, in the two characteristics the economy's exposure is written in (size, the proxy for the firm's type, and book-to-market, its leverage, in KP14 and BGN; profitability and book-to-market in GS21), with the benchmark in the top row and the 3b economy in the bottom row and one panel per method. The *interaction* figure evaluates each fitted surface on a grid over the two characteristics (the other three at their mean), subtracts the best additive fit $f(a) + g(b)$, and maps what is left: the part of the weight that cannot be written as an effect of one characteristic plus an effect of the other. White is no interaction, a saddle (two opposite corners red, two blue) is a product term: the slope in book-to-market differs by size. Each panel has its own colour scale, so shapes are compared across panels and sizes read from the standard deviation in the panel title, which is also the table's last column. The *line* figure plots the fitted weight against one characteristic at five levels of the other; there an interaction shows as fanning or crossing lines, mixed with the additive part the interaction figure removes. Fama–MacBeth is linear in the characteristics, so its panels are parallel lines and its interaction is noise; Fama–French's portfolios are double sorts on size and each characteristic, so it carries an interaction by construction, but in the shape of its sort breakpoints.

In every 3b economy the true weight is far from linear in the ranks (nonlinear share 0.39 to 0.58 of the surface), and the shape is an interaction: the weight's slope in one characteristic depends on the level of another. In KP14 (`vyg25`) the slope of the true weight in book-to-market depends on the firm's size (the proxy for its type) and on its momentum: steep for large and high-momentum firms, flat for the rest. Over size and book-to-market the interaction part of the true surface has a standard deviation of 0.106 and DKKM's 0.107, with the same saddle (the interaction figure); DKKM's nonlinear part correlates 0.77 with the truth's. Fama–MacBeth's interaction is 0.011, parallel lines by construction, and its nonlinear part correlates -0.13. Fama–French's interaction is 0.096, because its portfolios are size-by-characteristic double sorts, but its shape is the sorts' steps rather than the truth's saddle and its nonlinear part correlates 0.18. The truth's larger interaction, book-to-market with momentum (sd 0.148), DKKM does not recover (0.015): momentum is the noisier proxy for the type. In GS21 (`gvyp10`) 98% of the true weight is a function of the five characteristics and the surface

model	economy	weights	corr with truth	corr, MLP fit	nonlinear share	corr of nonlinear parts (t)	interaction sd at pair
BGN	`bgnbase` (benchmark; size × bm)	true SDF	1.00	0.77	0.35		0.076
		FMR	0.43	0.57	0.11	+0.28 (89.4)	0.012
		FF	0.35	0.44	0.35	+0.11 (6.8)	0.039
		DKKM	0.45	0.60	0.04	+0.37 (4.9)	0.008
BGN	`bvych0` (3b; size × bm)	true SDF	1.00	0.38	0.39		0.051
		FMR	0.17	0.21	0.07	+0.00 (0.4)	0.013
		FF	0.19	0.24	0.17	+0.19 (14.2)	0.019
		DKKM	0.24	0.24	0.03	+0.49 (59.9)	0.013
GS21	`gsbase` (benchmark; bm × roe)	true SDF	1.00	0.97	0.11		0.004
		FMR	0.54	0.74	0.07	+0.13 (14.3)	0.002
		FF	0.48	0.69	0.39	+0.30 (21.7)	0.002
		DKKM	0.57	0.79	0.14	-0.12 (-8.3)	0.000
GS21	`gvyp10` (3b; bm × roe)	true SDF	1.00	0.98	0.50		0.052
		FMR	0.25	0.28	0.16	+0.61 (68.8)	0.021
		FF	0.40	0.40	0.16	+0.33 (17.8)	0.025
		DKKM	0.83	0.86	0.38	+0.84 (83.9)	0.033
KP14	`kpbase` (benchmark; size × bm)	true SDF	1.00	0.80	0.08		0.008
		FMR	0.37	0.59	0.10	+0.02 (1.4)	0.006
		FF	0.31	0.42	0.20	+0.24 (17.3)	0.016
		DKKM	0.33	0.41	0.10	-0.10 (-23.5)	0.009
KP14	`vyg25` (3b; size × bm)	true SDF	1.00	0.67	0.58		0.112
		FMR	0.29	0.33	0.19	-0.13 (-18.2)	0.012
		FF	0.35	0.39	0.33	+0.18 (16.6)	0.095
		DKKM	0.48	0.55	0.21	+0.77 (229.0)	0.110

Table 7: Neural-network fits of the portfolio weights: benchmark and group-3b economies.
Columns: corr with truth = mean over (seed, month) cross-sections of the correlation between the method's weights and the true SDF weights; corr, MLP fit = the same for the MLP surface fitted to the method's weights; nonlinear share = the part of the fitted surface's variance that an OLS fit on the five ranks within each cross-section leaves (one minus the within-cross-section R^2); corr of nonlinear parts = cross-sectional correlation of that nonlinear remainder with the truth surface's nonlinear remainder, with its t-statistic across seeds; interaction sd at pair = standard deviation, in weight units, of the two-way interaction part of the fitted surface over the named pair (the surface minus its additive fit in the two characteristics), so the truth's row says how large the interaction is and a method's row how much of it the method's weights carry. The pair is the one the economy's exposure is written in (KP14 and BGN: type, proxied by size, times leverage, bm; GS21: productivity, roe, times leverage, bm) and is the pair the figures use. Seeds: three per benchmark, ten per 3b economy; evaluation months only; DKKM at $P = 3600$ and the economy's best κ; weights winsorized at 1/99 within cross-sections.

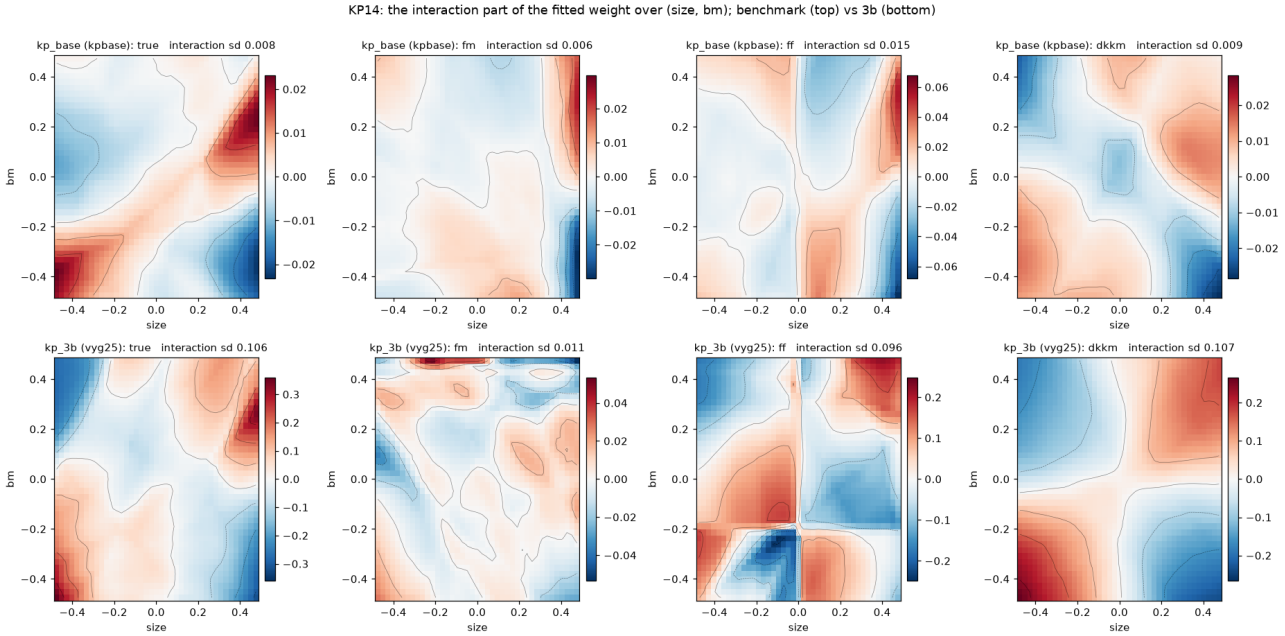


Figure 1: KP14: the interaction part of the fitted weight over book-to-market and size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

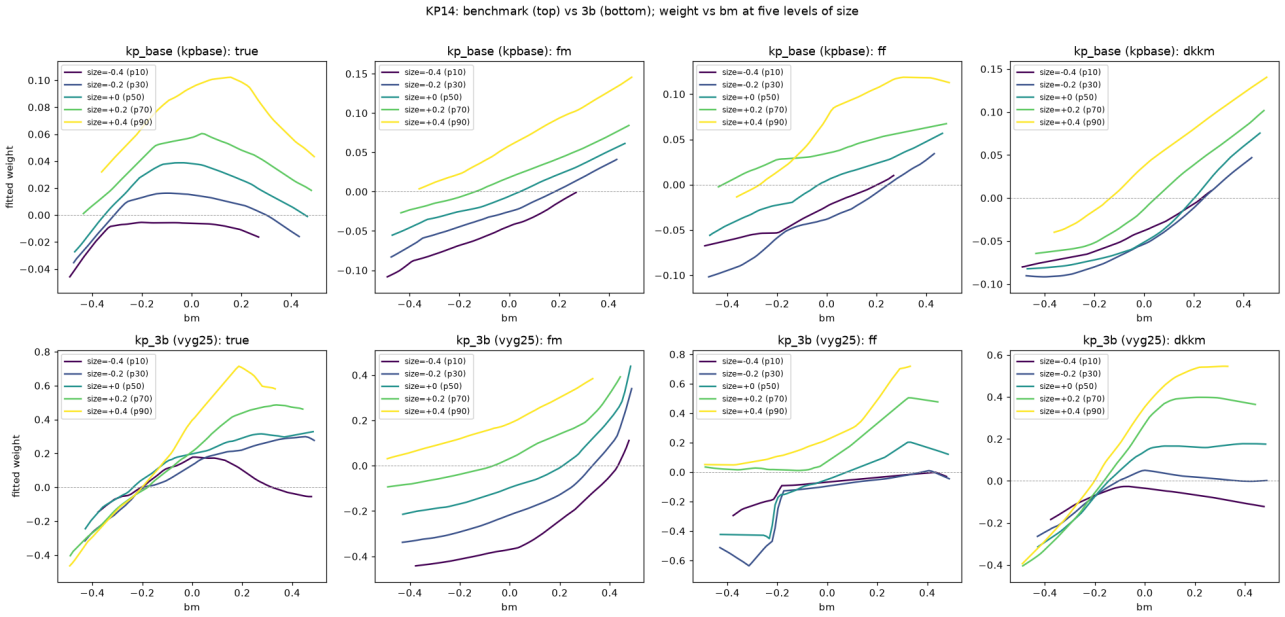


Figure 2: KP14: fitted weight against book-to-market at five levels of size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

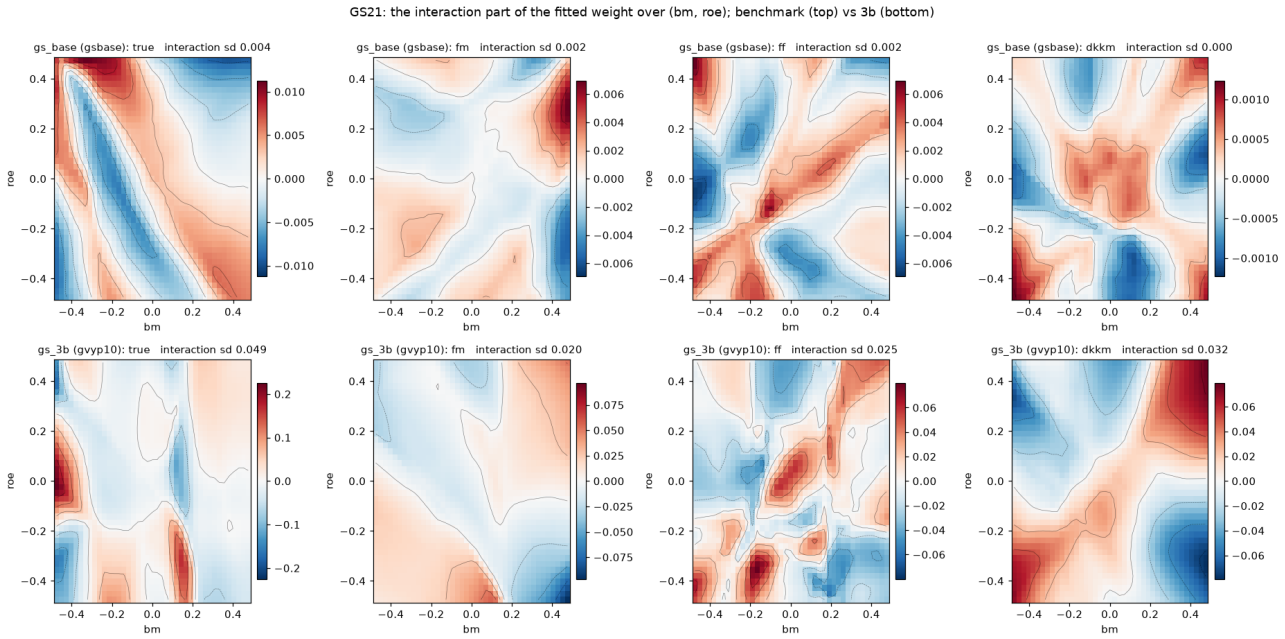


Figure 3: GS21: the interaction part of the fitted weight over profitability and book-to-market (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

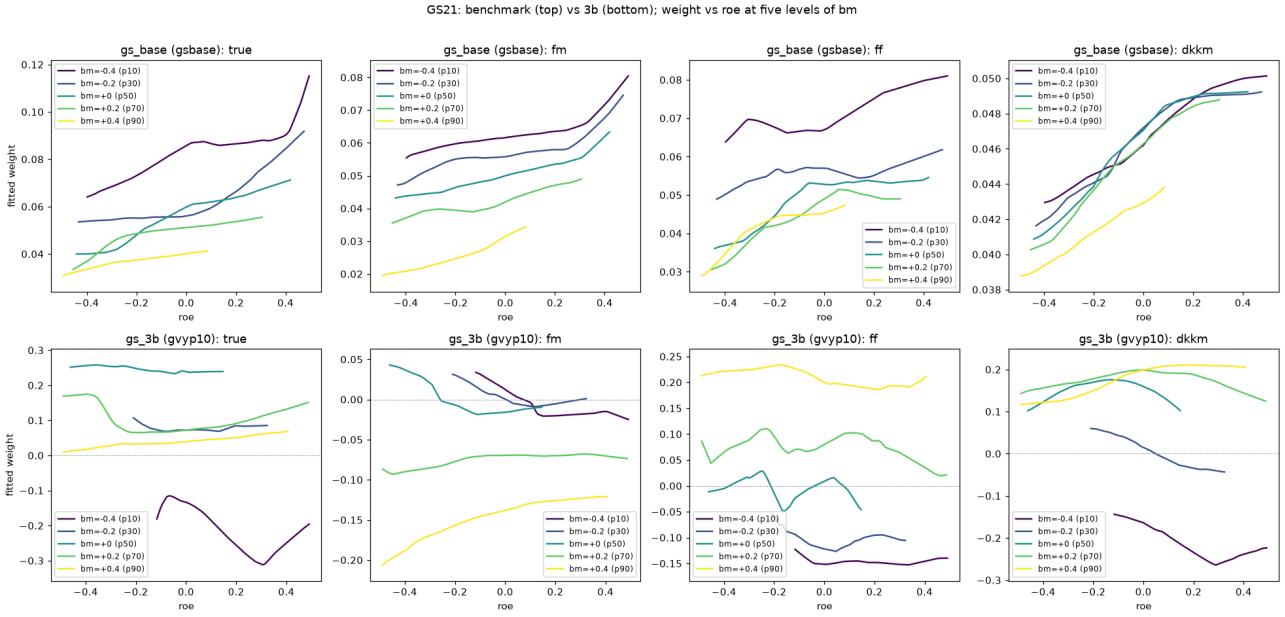


Figure 4: GS21: fitted weight against profitability at five levels of book-to-market (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

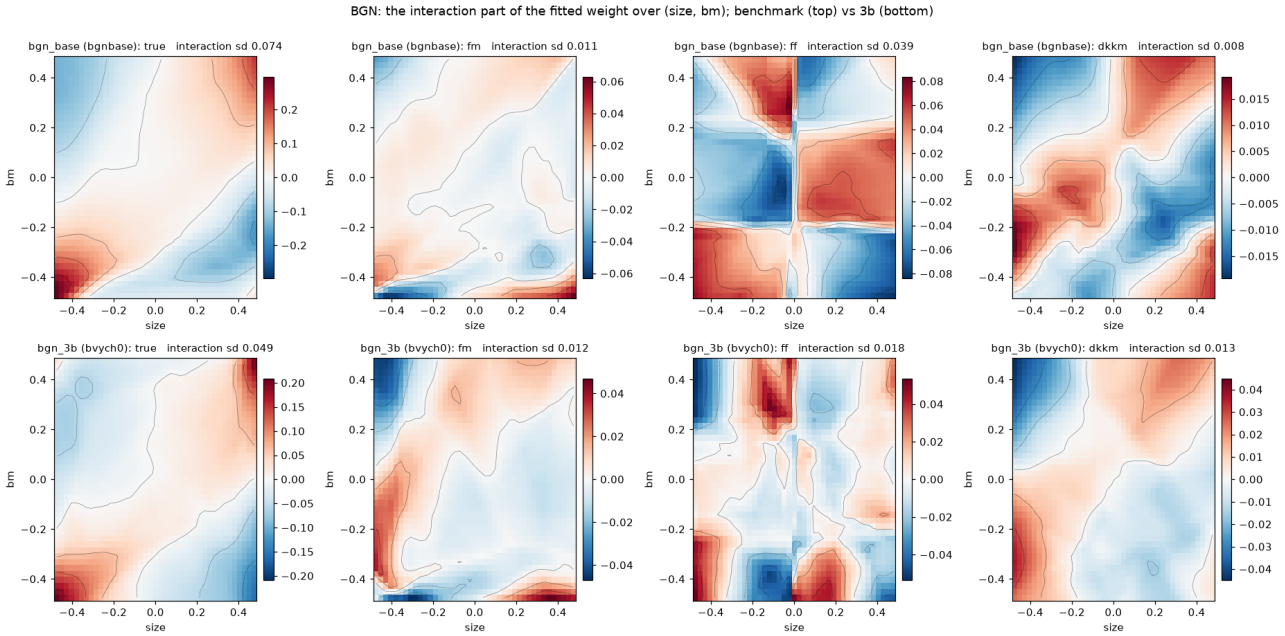


Figure 5: BGN: the interaction part of the fitted weight over book-to-market and size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM, each on its own colour scale, the interaction's sd in weight units in the title.

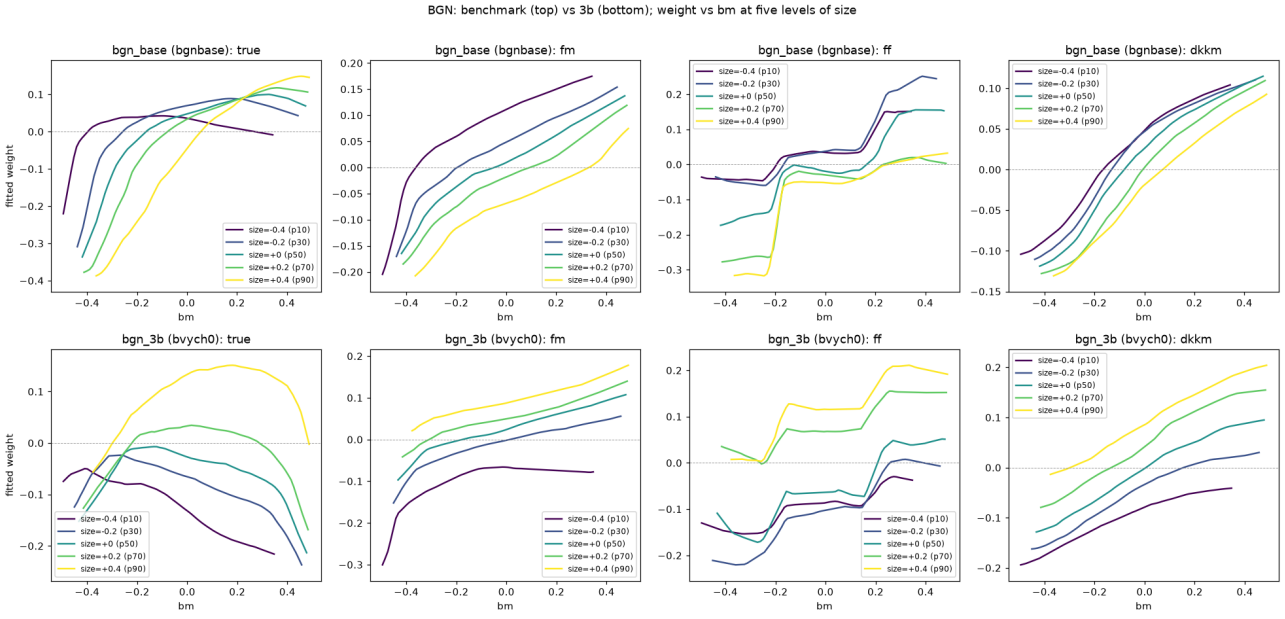


Figure 6: BGN: fitted weight against book-to-market at five levels of size (benchmark above, 3b below); panels: true SDF, Fama–MacBeth, Fama–French, DKKM; other characteristics at their mean.

is non-monotone in book-to-market: median-bm firms get the largest weight, the lowest-bm firms a negative one with a V in profitability. The DKKM weights reproduce this ordering (correlation with truth 0.83 for the weights themselves, 0.84 between the nonlinear parts); Fama–MacBeth gets the ordering of the bm levels wrong (0.25), and Fama–French (0.40) has the steps without the curvature. Over book-to-market and profitability the interaction's sd is 0.049 in the truth and 0.032 in DKKM, with the same pattern, against 0.020 and 0.025 for Fama–MacBeth and Fama–French with different patterns. In BGN (`bvych0`) only 38% of the true weight is a function of the five characteristics, because the exposure sits on a latent firm type that the characteristics proxy weakly; what is a function of them is a hump in book-to-market that grows with size. The DKKM surface is nearly linear here (nonlinear share 0.03): the size-by-book-to-market interaction has sd 0.049 in the truth and 0.013 in DKKM, the right saddle at a quarter of its size, and DKKM's nonlinear part correlates 0.49 with the truth's where Fama–MacBeth's is uncorrelated (0.00) and Fama–French's 0.19. The departure from linearity it captures is small, which is the Sharpe table's smallest 3b gap. The benchmarks are the control. In KP14 (`kpbase`) and GS21 (`gsbase`) the true weight is close to linear in the ranks (nonlinear share 0.08 and 0.11): parallel humps in book-to-market at every size in KP14, parallel rises in profitability at every book-to-market level in GS21, with nothing for a nonlinear method to find. The interaction over the figure pair has sd 0.008 and 0.004 in the truth and 0.009 and 0.000 in DKKM, and DKKM's nonlinear part is uncorrelated with the truth's (-0.10 and -0.12): what it adds to a linear portfolio is noise, which is the Sharpe tables' zero or negative gap, and in GS21 its best penalty is a hundred times the 3b economy's, shrinking it to the market. The BGN benchmark (`bgnbase`) is different: its true weight already has a size-by-book-to-market interaction (nonlinear share 0.35, interaction sd 0.074, the curves cross), and DKKM's surface is as linear there as in the 3b economy (nonlinear share 0.04, interaction sd 0.008); the BGN economies hand DKKM a nonlinearity it does not pick up in either version, which is why BGN's gaps are the smallest of the three models. Read across the figures, the mechanism is the same in all three models: DKKM's weights bend where the true weights bend, when the bend is an interaction of two characteristics large enough to show in the characteristics, and otherwise stay a linear portfolio plus noise.

7 What each economy changes

Notation. BGN (Berk, Green and Naik 1999): a firm is a portfolio of live projects; project j has systematic risk β_j, drawn from an exponential distribution, and lognormal cash flows C_j; it survives each month with probability π; a new project arrives each month and is taken if its NPV is positive; the pricing kernel prices the market shock at σ_z and the Vasicek short rate r_t. GS21 (Gomes and Schmid 2021): output is $e^{x_t+z_{it}}k_{it}$ with aggregate productivity x_t and idiosyncratic productivity z_{it} both AR(1); firms choose the coupon b_{it}, invest lumpily at a random cost, and may default; the kernel prices x-shocks at a constant $\gamma_x = 0.5$. KP14 (Kogan and Papanikolaou 2014): firm value is assets in place plus growth options; project cash flows are $x_t u_{it}\varepsilon_{jt}K_j^{\alpha}$, new projects arrive at the firm-specific rate λ_{it}, and the kernel prices disembodied productivity x at γ_x and embodied technology z at γ_z.

The added priced state y. Every economy with a y adds one stationary state with unit stationary variance and mean reversion $\kappa_y = 0.35$ per year, $y_{t+1} = \rho y_t + \sqrt{1-\rho^2}\,\eta_{t+1}$ with $\rho = e^{-\kappa_y/12}$ (an Ornstein–Uhlenbeck process in KP14, a Tauchen chain in GS21 and in BGN's chain economies). It is priced: the kernel is multiplied by $e^{-\lambda_y\eta_{t+1}-\lambda_y^2/2}$ with $\lambda_y = \gamma_v\sqrt{1/12}$, and $\gamma_v = 2.5$ unless stated. "Types" are equal shares of firms differing only in their exposure b_f to y. "Level-compensated" means type f's cash flows are also scaled by a constant $\theta_f = (V_0/V_f)^{1.2}$, the ratio of a unit project's value for type 0 and type f at $y = 0$, so that value at $y = 0$ does not reveal the type (KP14's ω device).

- **Baselines**
  - `bgnbase`: BGN as above, one type. Expected returns come from the value-weighted β_j of assets in place and from the growth option's exposure to r_t.
  - `gsbase`: GS21 as above, Table I calibration, one type.
  - `kpbase`: KP14 as above, one type, no y.
- **Group 1. A crash-only factor.** The 3a economy with the Gaussian price of y set to zero ($\gamma_v = 0$) and a crash.
  - `bvy1j`, `gvyz10j`, `vy1j`: `bvy1ch`, `gvyz10` and `vy1` with $\gamma_v = 0$ and, each month with probability 1% whatever y_t, a jump of y_{t+1} to -3.5 (the lowest node), weighted e^2 by the kernel (a risk-neutral probability of about 7% a month before renormalization). In KP14 the jump is a Poisson event at intensity 0.12 a year under P and $0.12\,e^2$ under Q.
- **Group 2. A quadratic price of the model's own risk.**
  - `bq25`: `bgnbase` with $\sigma_z \to \sigma_z\, g(r_t)$, $g(r) = \min(1 + 0.25\,((r - \bar{r})/\mathrm{sd}_r)^2, 3)$ binned to seven equiprobable bins of the stationary rate ($g = 1.00, 1.03, 1.17, 1.68$ from the middle outward); the date-k cash flow is priced at the bin of r_{t+k-1}, through seven bases $D_j(r)$ in place of the regime's two.
  - `gq25`: `gsbase` with $\gamma_x \to \gamma_x(1 + 0.25\,\tilde{x}^2)$ capped at 1.5, $\tilde{x} = x/\mathrm{sd}(x)$.
  - `kpq`: `kpbase` with $\gamma_x \to \gamma_x \min(1 + 0.25\,y^2, 3)$, y the unpriced stationary state of `kpgam` ($\kappa = 0.35$, unit variance); no firm is exposed to y.
- **Group 3a. One type exposed to y.**
  - `bvy1ch`: `bgnbase` plus y (21-node chain), one type; project cash flows $C_j e^{0.035\,y_t}$.
  - `gvyz10`: `gsbase` plus y (9-node chain over ± 3.5 sd), one type; output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\,y_t} k_{it}$, $\tilde{z} = z/\mathrm{sd}(z)$: the product economy's exposure without the coupon.
  - `vy1`: `kpbase` plus y, one type; project cash flows $e^{0.07\,y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$.
- **Group 3b. Interactions.**
  - `bvych0`: `bvy1ch` with three types, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated ($\theta_f = (V_0/V_f)^{1.2}$). Exposure is b_f times the firm's mix of assets in place and growth option.
  - `gvyp10`: `gsbase` plus y (9-node chain), one type, output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$, with $\tilde{z} = z/\mathrm{sd}(z)$ and $\tilde{b} = (b - 0.5)/0.29$ the standardized coupon. Exposure is productivity $\times$ leverage, revealed by profitability and leverage.
  - `vyg25`: `vy1` with three types, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated. Growth options are options on the $e^{b_f y}$ cash flows, so a firm's exposure is b_f times the elasticity of its value, which rises with its growth-option share: type $\times$ growth-option share, revealed by size and book-to-market.
- **Group 4. Two priced states.** Total price fixed, $\gamma_v = 2.5/\sqrt{2}$ on each.
  - `bvyk2`: `bvych0` with a second independent 21-node chain $y^{(2)}$; project cash flows

$$C_j\, e^{b_f y_t + b_f^{(2)} y_t^{(2)}}, \qquad (b_f, b_f^{(2)}) \in \{(0.01, 0),\ (0, 0.035),\ (0.049, 0.049)\}.$$

- `gvypk3`: `gvyp10` with a second independent chain (5 nodes each, 25 joint) and three types of equal share, loading $0.10\,\tilde{z}\tilde{b}$ on y, on $y^{(2)}$, and $0.071\,\tilde{z}\tilde{b}$ on both.
- `vy9t2`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; a firm's type is a pair (f_1, f_2) drawn independently from $\{0.02, 0.07, 0.14\}$ for each state, nine types of equal share, and its cash flows load $e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each state.

Reduced form (Table 6)

A simulated panel (`variants/reduced/rf_model.py`) of $N = 500$ firms over $T = 500$ months after a 100-month burn-in, five seeds per cell, run through the same oracle ceilings and the same estimators as the structural economies at the 360-month window; its true conditional moments are known in closed form. Notation: i indexes firms, t months, $j = 1, \ldots, 5$ the characteristics (size, book-to-market, asset growth, profitability, momentum), $k = 1, \ldots, K$ the extra priced factors.

- **Latent processes.** Each firm carries five independent AR(1) processes $z_{ijt} = \rho_j z_{ij,t-1} + \sqrt{1-\rho_j^2}\,\eta_{ijt}$, η standard normal, $\rho = (0.99, 0.98, 0.90, 0.95, 0.70)$, so each has unit stationary variance.
- **Observed characteristics.** $X_{ijt} = z_{ijt}$ in the `kfh` and `kfo` ladders. In the `lfh` ladder the five processes are the firm's true states and each characteristic is its state plus observation noise, $X_{ijt} = z_{ijt} + \sigma_\nu \nu_{ijt}$, with ν_{ijt} an independent AR(1) with the same ρ_j and unit variance. The ladder's one noise parameter is σ_ν: 0 (the states are observed exactly, rows `n0`) or 1 (noise as large as the state itself, rows `n1`). $\tilde{X}_{it}$ is X_{it} standardized across firms at t. The oracle's bases and the estimators see the ranks of X_{it}, as everywhere in the report.
- **Returns.** $r_{i,t+1} = B_{it}\lambda + B_{it} f_{t+1} + \varepsilon_{i,t+1}$, with $f_{t+1} \sim N(0, \sigma_f^2 I_{K+1})$, $\sigma_f = 0.04$, and $\varepsilon_{i,t+1} \sim N(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon = 0.10$, all independent. B_{it} is the $1 \times (K+1)$ row of firm i's loadings and λ the $(K+1)$-vector of constant prices of risk. The true premium is $\mu_{it} = B_{it}\lambda$ and the true covariance $\Sigma_t = \sigma_f^2 B_t B_t' + \sigma_\varepsilon^2 I$.
- **The market factor (column 0).** $B_{i0t} = 1 + a'\tilde{X}_{it}$ with $a = (0, 0.3, 0, 0.3, 0.3)$ on (size, bm, agr, roe, mom), and $\lambda_0 = 0.002$, a Sharpe of $\lambda_0/\sigma_f = 0.05$. This loading is linear in the characteristics, so every method holds it.
- **The K extra factors (columns $k = 1, \ldots, K$).** $\lambda_k = 0.024/\sqrt{K}$ for every k, so each has Sharpe $0.6/\sqrt{K}$ and the total price of risk is the same at every K. Every extra loading is centered across firms each month ($\bar{\beta}_{kt}$ is the cross-sectional mean), so the equal-weighted market carries none of the extra premia. The ladders differ only in the loading:
  - ***kfh*** (loadings on observed characteristics): $B_{ikt} = h(w_k'\tilde{X}_{it}) - \bar{\beta}_{kt}$, with w_k a fixed unit vector in $\mathbb{R}^5$ drawn once and reused in every seed, and $h(u) = u$ in the main table (lin); the appendix adds $h(u) = \tanh(u)$ (tanh) and $h(u) = u^2$ (sq). Exposure is a function of what the characteristics show directly: GS21's structure.
  - ***kfo*** (appendix): `kfh` with the size coordinate of each w_k set to zero before normalizing, so no extra factor loads on size.
  - ***lfh*** (loadings on products of latent states): $B_{ikt} = z_{ia_kt}\, z_{ib_kt} - \bar{\beta}_{kt}$, where (a_k, b_k) is the k-th pair in a fixed enumeration of the ten pairs of the five processes (so $K \le 10$). The loading is a product of two true states that the characteristics reveal only up to the observation noise σ_ν: KP14's and BGN's structure, a type times a within-firm state.

The ladders therefore vary one thing at a time: K; whether the loading is a function of the observed characteristics or a product of partly hidden states; and, within `lfh`, how much noise hides the states. In the `kfh` linear and tanh ladders $\mathrm{SR}_{\max}$ falls at $K = 16$ because sixteen directions over five characteristics are collinear and their loadings partly cancel; in the `lfh` ladder $\mathrm{SR}_{\max}$ stays near 0.6 at every K.

8 Additional economies

Everything else that was run. The table has the main tables' columns; the bullets describe each economy against its parent (notation as in Section 7). Three groups of them answer side questions: the crash added to the interaction economies instead of to a single-type economy (`bvyj25`, `gvyp10j`, `vyj25`), the GS21 product ladder (`gvyp05` to `gvyp20`, `gvyp10x`), and the earlier single-characteristic and price-of-risk wirings that the harmonized cells replace.

Table 8: Additional economies. Everything else that was run, by model, each described in the appendix against its parent; the RF rows are the reduced-form ladders with bounded (tanh), curvature (square) and size-orthogonal curvature (kfo) loadings.

model	economy	$\mathrm{SR}_{\max}$	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`g0235f` – regime price ×0.2 / ×3.5, spells 4× faster	0.237	62%	0.163	+0.045	28	0.13	0.169	0.141	0.131	0.161	0.146	+0.0079 (t 4.8)	+0.05	10/10
BGN	`g0235df` – stress 2/3 of the time, 4× faster	0.489	17%	0.432	+0.036	9	0.18	0.423	0.392	0.351	0.428	0.077	-0.0051 (t -2.2)	-0.05	2/10
BGN	`bvy25` – three types on y, level-compensated (1.2)	0.601	41%	0.398	+0.062	16	0.23	0.399	0.344	0.367	0.384	0.248	+0.0148 (t 3.0)	+0.09	9/10
BGN	`bvych41` – bvych0 on a 41-node chain	0.565	38%	0.336	+0.052	15	0.19	0.328	0.279	0.291	0.320	0.215	+0.0086 (t 2.8)	+0.06	8/10
BGN	`bvyc0` – the same without compensation	0.475	41%	0.395	+0.066	18	0.22	0.386	0.321	0.347	0.376	0.195	+0.0104 (t 3.1)	+0.07	9/10
BGN	`bvyj25` – the chain with a priced crash (1%/month to $-y_{\max}$, priced e^2)	0.536	38%	0.315	+0.054	17	0.19	0.307	0.261	0.269	0.301	0.202	+0.0066 (t 2.8)	+0.05	9/10
BGN	`bvysg` – signed exposure ladder (-0.07 / 0 / $+0.07$)	0.504	28%	0.361	+0.120	35	0.31	0.318	0.305	0.273	0.324	0.141	-0.0199 (t -2.0)	-0.08	2/10
GS21	`g28` – countercyclical $\gamma(x)$, clipped	0.314	95%	0.312	+0.001	0	0.02	0.303	0.254	0.256	0.305	0.298	-0.0024 (t -4.3)	-0.04	0/10
GS21	`bx7` – two-state regime on γ, five types on x	0.327	94%	0.317	+0.007	2	0.07	0.309	0.282	0.271	0.311	0.306	-0.0017 (t -2.8)	-0.03	0/10
GS21	`gx7` – five types on productivity x	0.485	98%	0.474	+0.008	2	0.08	0.473	0.393	0.383	0.473	0.473	+0.0003 (t 1.6)	+0.01	5/10
GS21	`gvy1` – one type, output $\times e^{0.05y}$	0.636	95%	0.633	+0.002	0	0.05	0.625	0.613	0.601	0.629	0.602	-0.0036 (t -4.2)	-0.06	1/10
GS21	`gvy1j` – crash-only factor: $e^{0.05y}$, the same crash	0.379	96%	0.377	+0.001	0	0.03	0.372	0.353	0.343	0.374	0.363	-0.0027 (t -6.1)	-0.04	0/10
GS21	`gvy25` – production-level exposure to y, three types	0.725	86%	0.709	+0.014	2	0.14	0.686	0.681	0.670	0.690	0.625	-0.0041 (t -3.5)	-0.06	2/10
GS21	`gvys25` – signed fixed-cost exposure to y	0.619	69%	0.602	+0.014	2	0.13	0.496	0.490	0.426	0.499	0.420	-0.0032 (t -1.5)	-0.03	2/6
GS21	`gvyp05` – product $\bar{z}\bar{b}$ at 0.05	0.425	60%	0.284	+0.128	45	0.30	0.296	0.255	0.206	0.277	0.254	+0.0194 (t 6.4)	+0.10	10/10
GS21	`gvyp075` – product at 0.075	0.561	28%	0.261	+0.287	111	0.48	0.428	0.172	0.188	0.351	0.149	+0.0765 (t 8.0)	+0.24	10/10
GS21	`gvyp15` – product at 0.15	0.744	-25%	0.616	+0.119	22	0.39	0.661	0.327	0.515	0.583	0.192	+0.0780 (t 6.1)	+0.30	10/10
GS21	`gvyp20` – product at 0.20	0.778	-51%	0.729	+0.039	6	0.23	0.701	0.452	0.643	0.680	0.407	+0.0215 (t 2.8)	+0.14	9/10
GS21	`gvyp10x` – product at 0.10 plus a curvature loading on x	0.589	17%	0.365	+0.215	63	0.45	0.483	0.157	0.258	0.435	0.054	+0.0486 (t 7.0)	+0.21	10/10
GS21	`gvyp10j` – the product exposure with a priced crash	0.537	21%	0.328	+0.197	62	0.41	0.440	0.210	0.240	0.409	0.072	+0.0301 (t 4.7)	+0.15	10/10
GS21	`gvypk2` – two priced chains, two product types	0.478	27%	0.226	+0.212	103	0.37	0.305	0.143	0.156	0.254	0.105	+0.0510 (t 2.9)	+0.13	9/10
GS21	`gvyzl` – productivity exposure, three fixed-coupon leverage types (diluting)	0.865	50%	0.793	+0.059	8	0.31	0.804	0.797	0.746	0.798	0.426	+0.0042 (t 2.0)	+0.05	7/10
GS21	`gvyzlf` – productivity exposure, three leverage types, capital structure fixed for life	0.880	50%	0.808	+0.066	8	0.33	0.820	0.810	0.760	0.815	0.432	+0.0045 (t 1.8)	+0.05	7/10
GS21	`gvyzlw` – gvyzlf with coupons 0.05 / 0.45 / 0.85	0.782	52%	0.729	+0.048	7	0.27	0.723	0.733	0.677	0.739	0.404	-0.0178 (t -5.4)	-0.16	0/10
GS21	`gvyzl5` – gvyzlf with five leverage types	0.880	50%	0.812	+0.060	7	0.32	0.819	0.817	0.767	0.818	0.440	-0.0017 (t -0.6)	-0.02	5/10
GS21	`gvyzlb` – gvyzlf with the exposure doubled	1.066	45%	1.003	+0.056	7	0.31	0.993	0.972	0.947	0.983	0.490	+0.0092 (t 1.7)	+0.07	7/10
GS21	`gvyzlr` – gvyzlf with riskless debt	0.880	49%	0.808	+0.066	8	0.33	0.820	0.810	0.760	0.815	0.432	+0.0044 (t 1.8)	+0.05	7/10
GS21	`gvyzlp` – gvyzlf with a persistent y (κ_y 0.10)	0.475	76%	0.457	+0.017	4	0.13	0.445	0.436	0.423	0.452	0.355	-0.0066 (t -3.5)	-0.06	2/10
GS21	`gvyzlrp` – gvyzlf with riskless debt and a persistent y	0.475	76%	0.457	+0.017	4	0.13	0.446	0.436	0.423	0.452	0.356	-0.0066 (t -3.5)	-0.06	2/10
GS21	`gsbg1` – benchmark at γ_x 0.1	0.070	96%	0.070	+0.000	0	0.00	0.061	0.031	0.030	0.061	0.054	-0.0000 (t -0.3)	-0.00	5/10
GS21	`gvyz10g` – gvyzl0 at γ_x 0.1	0.770	25%	0.707	+0.051	7	0.27	0.704	0.725	0.648	0.725	0.195	-0.0242 (t -5.7)	-0.18	0/10
GS21	`gvyzlg` – gvyzlf at γ_x 0.1	0.789	27%	0.718	+0.064	9	0.31	0.725	0.727	0.664	0.728	0.217	-0.0057 (t -2.0)	-0.07	2/10
GS21	`gvyp10c` – the product with a leverage-level term, exposure 0.10 $\bar{b}(\bar{z}+2)$	0.774	-86%	0.763	+0.006	1	0.09	0.745	0.727	0.726	0.763	0.674	-0.0180 (t -4.9)	-0.16	0/10
GS21	`gvyt1` – three exposure types × three fixed leverage types (KP14's recipe)	0.739	84%	0.728	+0.009	1	0.12	0.706	0.687	0.687	0.706	0.621	-0.0004 (t -0.4)	-0.00	5/10
GS21	`gvyt1g` – the same at γ_x 0.1	0.517	72%	0.498	+0.017	3	0.13	0.467	0.445	0.446	0.465	0.374	+0.0020 (t 1.3)	+0.02	6/10
BGN	`bvy9t2` – two priced chains, nine types (independent ladders), total price 2.5	0.636	42%	0.398	+0.063	16	0.23	0.395	0.339	0.355	0.383	0.266	+0.0124 (t 3.5)	+0.08	8/10
KP14	`kpgam` – $\gamma_x \times$ logistic(y), 1 to 3	0.390	65%	0.373	+0.005	1	0.06	0.358	0.348	0.340	0.363	0.254	-0.0053 (t -4.0)	-0.05	1/10
KP14	`vyx` – three types on y, price 1.8	0.421	14%	0.321	+0.055	17	0.19	0.313	0.284	0.302	0.310	0.054	+0.0009 (t 0.3)	+0.02	7/10
KP14	`vyj25` – the same with a priced crash in y	0.554	10%	0.323	+0.104	33	0.28	0.315	0.270	0.292	0.307	0.045	+0.0080 (t 2.2)	+0.06	8/10
KP14	`vym2t3` – two priced states, three loading vectors	0.475	12%	0.326	+0.090	28	0.26	0.317	0.280	0.301	0.308	0.049	+0.0075 (t 2.1)	+0.05	8/10
KP14	`vy9t2f` – two priced states, nine types, each state priced 2.5	0.582	6%	0.415	+0.077	19	0.26	0.416	0.314	0.330	0.401	0.018	+0.0149 (t 4.2)	+0.10	9/10
KP14	`vym3t3` – three priced states, three types	0.443	14%	0.285	+0.087	30	0.24	0.279	0.246	0.266	0.273	0.061	+0.0036 (t 0.8)	+0.02	6/10
KP14	`vym3` – three priced states, twenty rank-3 types	0.418	15%	0.298	+0.056	19	0.19	0.286	0.259	0.278	0.287	0.059	-0.0032 (t -1.7)	-0.02	5/10
RF	`kfh_tanh_1` – K=1, bounded (tanh) loadings on observed characteristics	0.592	8%	0.592	+0.001	0	0.03	0.579	0.578	0.570	0.588	0.041	-0.0105 (t -1.9)	-0.10	0/5
RF	`kfh_tanh_4` – K=4	0.584	9%	0.574	+0.007	1	0.09	0.541	0.564	0.544	0.560	0.041	-0.0232 (t -2.9)	-0.16	0/5
RF	`kfh_tanh_16` – K=16	0.469	11%	0.378	+0.044	12	0.19	0.346	0.357	0.345	0.359	0.041	-0.0152 (t -1.4)	-0.10	1/5
RF	`kfh_sq_1` – K=1, pure curvature (squares) of observed characteristics	0.600	8%	0.080	+0.519	647	0.59	0.582	0.032	0.046	0.059	0.041	+0.5161 (t 57.3)	+0.58	5/5
RF	`kfh_sq_4` – K=4	0.601	8%	0.097	+0.501	515	0.59	0.573	0.042	0.458	0.074	0.041	+0.1155 (t 22.6)	+0.35	5/5
RF	`kfh_sq_16` – K=16	0.601	8%	0.076	+0.523	687	0.59	0.571	0.036	0.429	0.052	0.041	+0.1429 (t 14.3)	+0.38	5/5
RF	`kfo_sq_1` – K=1, squares, directions orthogonal to size	0.600	8%	0.080	+0.519	648	0.59	0.582	0.032	0.048	0.059	0.041	+0.5163 (t 57.7)	+0.58	5/5
RF	`kfo_sq_4` – K=4	0.601	8%	0.092	+0.507	552	0.59	0.575	0.058	0.061	0.081	0.041	+0.4833 (t 25.4)	+0.57	5/5
RF	`kfo_sq_16` – K=16	0.602	8%	0.079	+0.520	656	0.59	0.575	0.015	0.031	0.053	0.041	+0.5030 (t 21.3)	+0.57	5/5

BGN

- `g0235f`: `bgnbase` with $\sigma_z \to g_s\sigma_z$, $s_t \in \{\text{calm}, \text{stress}\}$ a Markov chain, $g = 0.2$ calm and 3.5 stress; monthly switch probabilities 1/12 (calm to stress) and 1/6 (stress to calm), so stress is one third of months with a mean spell of 6 months.
- `g0235df`: the same g with switch probabilities 1/6 and 1/12: stress two thirds of months, mean spell 12 months.
- `bvy25`: `bgnbase` plus y; type f's project cash flows are $\theta_f C_f e^{b_f y_t}$, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated. Exposure is b_f times the firm's mix of assets in place and growth option.
- `bvych41`: `bvych0` with y on a 41-node chain instead of 21.

- `bvy9t2`: `bvych0` with a second independent 21-node chain and nine types, a pair (f_1, f_2) drawn independently from $\{0.01, 0.035, 0.07\}$ for each chain, cash flows $C_j\, e^{b_{f_1} y_t + b_{f_2} y_t^{(2)}}$; price $2.5/\sqrt{2}$ on each: the nested nine-type design of `vy9t2` in BGN.

- `bvyc0`: `bvy25` with $\theta_f = 1$, so the type is visible in value.

- `bvyj25`: `bvych0` (group 3b) plus a crash: each month, with probability 1% whatever y_t, y_{t+1} goes to the lowest node; otherwise the Tauchen transition. The kernel weights the crash transition by e^2, so its risk-neutral probability is about 7.4% a month before renormalization.

- `bvysg`: `bgnbase` plus y; type f's project cash flows are $C_j e^{b_f y_t}$, $b_f \in \{-0.07, 0, +0.07\}$, not level-compensated.

GS21

- `g28`: `gsbase` with $\gamma_x \to \gamma(x_t) = \text{clip}\big(0.5 - 0.28\, x_t/\text{sd}(x),\, 0.05,\, 1.0\big)$, countercyclical.

- `bx7`: `gsbase` with $\gamma_x \to g_s \gamma_x$, $g \in \{0.6, 3.0\}$ on a two-state Markov regime, and five types with output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$.

- `gx7`: `gsbase` with five types, output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$, and $\gamma(x_t) = \text{clip}\big(0.5 - 0.4\, x_t/\text{sd}(x),\, 0.05,\, 1.5\big)$. A firm's exposure to the one priced shock is its type, which size and profitability reveal.

- `gvy1`: `gsbase` plus y (9-node chain over ± 2.5 sd), one type; output $e^{x_t + z_{it} + 0.05\, y_t} k_{it}$, a level exposure.

- `gvy1j`: `gvy1` with $\gamma_v = 0$ and the crash, on a chain over ± 3.5 sd.

- `gvy25`: `gsbase` plus y (9-node chain); output $e^{x_t + z_{it} + b_f y_t + a_f} k_{it}$, $b_f \in \{0.015, 0.05, 0.10\}$, with a_f the level compensation.

- `gvys25`: `gsbase` plus y, entering through an operating cost: $0.2\, e^{c_f y_t}$ per unit capital is subtracted from output, $c_f \in \{+0.4, 0, -0.4\}$; output itself does not load on y.

- `gvyp05, gvyp075, gvyp15, gvyp20`: `gvyp10` with the coefficient 0.10 replaced by 0.05, 0.075, 0.15 and 0.20.

- `gvyp10x`: `gvyp10` with output $e^{(1+(\tilde{z}_{it}^2 - 1)/\sqrt{2})\, x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t} k_{it}$: the loading on x is also nonlinear in productivity.

- `gvyp10j`: `gvyp10` (group 3b) plus the same priced crash on its 9-node chain.

- `gvypk2`: `gvyp10` with a second independent chain (5 nodes each, 25 joint); two types of equal share with output $e^{x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t} k_{it}$ and $e^{x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t^{(2)}} k_{it}$.

- `gvyzl`: `gvyz10` with three leverage types: the coupon the firm refinances to is fixed at 0.158 / 0.421 / 0.684 per unit capital, equal shares drawn independently of productivity; refinancing opportunities (1% a month) and dilution at investment as in GS21, so the realized coupon drifts below the type's.

- `gvyzlf`: `gvyzl` with the capital structure fixed for life: no refinancing, new debt issued pari passu at investment (debt-to-equity 0.25 / 1.0 / 3.25).

- `gvyzlw`: `gvyzlf` with coupons 0.05 / 0.45 / 0.85; the high type is distressed (premium 16% a year, default rate 1.4%).
- `gvyzl5`: `gvyzlf` with five types, coupons 0.10 to 0.70.
- `gvyzlb`: `gvyzlf` with the exposure doubled to $0.20\,\tilde{z}\,y$.
- `gvyzlr`: `gvyzlf` with riskless debt: the debt claim is worth its riskless coupon value in every state (creditors made whole on default).
- `gvyzlp`: `gvyzlf` with $\kappa_y = 0.10$ a year instead of 0.35.
- `gvyzlrp`: both.
- `gsbg1`: the GS21 benchmark with $\gamma_x = 0.1$ instead of 0.5 (`gs_vy` at one y node and no exposure, which is `gsbase`'s solver).
- `gvyz10g`: `gvyz10` at $\gamma_x = 0.1$.
- `gvyzlg`: `gvyzlf` at $\gamma_x = 0.1$.
- `gvyp10c`: `gvyp10` with a leverage-level term, output $e^{x+z+0.10\,\tilde{b}(\tilde{z}+2)\,y}$: a levered firm's exposure keeps leverage's sign for every firm with $\tilde{z} > -2$. The mean firm ($\tilde{b} < 0$) is a hedge against y, so the market's premium is negative.
- `gvytl`: KP14's 3b recipe in GS21: three latent exposure types, output $e^{x+z+b_f y+a_f}$ with $b_f \in \{0.015, 0.05, 0.10\}$ and a_f the level compensation, times three fixed leverage types (coupons 0.158 / 0.421 / 0.684, no refinancing), nine types of equal share.
- `gvytlg`: `gvytl` at $\gamma_x = 0.1$ (benchmark `gsbg1`).

KP14

- `kpgam`: `kpbase` with $\gamma_x \to \gamma_x\, g(y_t)$, $g(y) = 1 + 2/(1 + e^{2y})$, rising from 1 when y is high to 3 when it is low; y is unpriced ($\gamma_v = 0$) and no firm is exposed to it.
- `vyx`: `kpbase` plus y with $\gamma_v = 1.8$; type f's project cash flows are $\theta_f e^{b_f y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated.
- `vyj25`: `vyg25` (group 3b) plus a Poisson jump of y to $-y_{\max} = -3.5$ at intensity 0.12 a year (1% a month) under P and $0.12\,e^2 = 0.89$ a year under Q.
- `vym3t3`: `vyg25` with $y \in \mathbb{R}^3$, three independent OU states; type f's cash flows load $e^{b_f' y_t}$ with three vectors $b_f \in \mathbb{R}^3$ spanning the three states.
- `vym2t3`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; three types with loading vectors $(0.02, 0)$, $(0, 0.07)$, $(0.099, 0.099)$, price $2.5/\sqrt{2}$ on each: three sorts in three directions, which prices types 1 and 2 at an effective 1.77 instead of 2.5. The group-4 cell until October 21, 2026.
- `vy9t2f`: `vy9t2` with each state priced at 2.5 instead of $2.5/\sqrt{2}$; nests `vyg25` exactly.
- `vym3`: the same three states with twenty types, $b_f \in \mathbb{R}^3$ of rank 3.

What they add to the main results:

- **The chain's resolution is not what separates `bvych0` from `bvy25`**: on 41 nodes the gap is +0.009 against +0.008 on 21, and the difference from the Gaussian economy's +0.015 (sd 0.016) is within noise.

- **A level exposure of GS21 output to y** (`gvy1`, `gvy1j`) has no room at all, as every other level wiring in GS21; the productivity exposure (`gvyz10`) has 5% and the product with the coupon 43%.

- **The crash on an interaction economy** leaves room and gap at the control's in BGN (+0.007 against +0.008) and GS21 (+0.030 against +0.020, within noise) and halves KP14's (+0.008 against +0.014): the crash months are a state the ranks-times-state linear benchmark reads as well as DKKM does.

- **The GS21 product ladder**: room is hump-shaped in the product's size (45 / 111 / 47 / 22 / 6% of the linear ceiling at 0.05 / 0.075 / 0.10 / 0.15 / 0.20) while the gap is +0.019 / +0.077 / +0.020 / +0.078 / +0.022; above 0.10 the market becomes a hedge. A second nonlinear loading on x (`gvyp10x`) gives +0.049: a second interaction factor adds. `bvy25`, the Gaussian-y version of `bvych0`, has twice its gap at the same room, and the visibility ladder (`bvy25`, `bvyc0`, and `bvyc10` at `bv_comp` 1.0: +0.015 / +0.010 / +0.010) is flat within noise.

- **Two apparent failures of the mechanism are estimation cost, not structure.** `vyx` is `vyg25` at a lower price of y: its room (0.054) is about what DKKM gives up to estimation relative to the linear model (0.053), so the gap is zero, and the higher price lifts the room more than the cost. `bvysg` has twice `bvy25`'s room but DKKM's estimation loss is two and a half times larger (0.16 against 0.06): with signed exposures the nonlinear part of the tangency portfolio is a direct bet on the persistent state y, whose sample mean a 360-month window barely pins down. What matters is room net of DKKM's estimation cost. Four more GS21 wirings (investment cost, floating coupon, a larger option, state-dependent recovery) were stopped at the oracle stage with room below +0.01, and GS21's default channel is dormant at every calibration tried.

- **Knobs that are not mechanisms**: the speed of the priced state (κ 0.15 / 0.35 / 0.70 in KP14: +0.007 / +0.014 / +0.009, a mild hump within noise), the visibility of the type, the number of types (`vym3` against `vym3t3`), the sample size (`vym3` at $N = 1000$: room up, gap flat), and the ridge grid (DKKM +0.0001). The price of the interacted component is the one scaling lever (`vyx` to `vyg25`).

Two priced states

A second independent priced state, with the total price held at the one-state value ($2.5/\sqrt{2}$ on each). In BGN and GS21 three types span the two states with loading vectors in the pattern $(1, 0)$, $(0, 1)$, $(1, 1)/\sqrt{2}$, at the 3b exposure sizes; in GS21 each type carries the 0.10 product exposure to its state. In KP14 the cell is the nested extension of 3b: every firm draws a type for each state independently from the 3b ladder $(0.02, 0.07, 0.14)$, nine types in all, so exposure to each state is its own type times leverage and zeroing the second ladder recovers `vyg25`. The three-direction KP14 version (`vym2t3`), the nested version with each state priced at the full 2.5 (`vy9t2f`) and the nested BGN version (`bvy9t2`) are in the appendix.

Verdict. More factors are a lever only where the second state brings a second interaction. In BGN the second chain raises room by a quarter and doubles the gap (+0.015, ten of ten seeds). In GS21 it raises the product economy's gap from +0.020 to +0.056 (nineteen of twenty seeds), as the two-type version did (appendix). In KP14 it does not help however it is wired: the nested nine-type cell lowers

Table 9: Group 4. Larger factor models: two priced states. Two independent priced states at total price 2.5 ($2.5/\sqrt{2}$ each). BGN and GS21: three types with loading vectors (1,0), (0,1), $(1,1)/\sqrt{2}$. KP14: nine types, an independent draw from the 3b ladder for each state (the three-direction version and the full-price version are in the appendix).

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	interaction SR	DKKM	FMR	FF	linear	CAPM	fair gap	DKKM ⊥	+
BGN	`bvyk2` – two priced chains, three loading vectors	0.639	36%	0.393	+0.071	18	0.25	0.394	0.332	0.353	0.379	0.228	+0.0152 (t 4.3)	+0.10	10/10
GS21	`gvypk3` – two priced chains, three product types	0.612	8%	0.349	+0.226	65	0.45	0.465	0.245	0.306	0.407	0.050	+0.0563 (t 4.7)	+0.20	19/20
KP14	`vy9t2` – two priced states, nine types (independent ladders), total price 2.5	0.506	9%	0.387	+0.056	15	0.21	0.378	0.326	0.337	0.373	0.031	+0.0048 (t 2.3)	+0.04	7/10

room from 24% to 15% of the linear ceiling and the gap from +0.015 to +0.005; the three-direction version (`vym2t3`) keeps the room (28%) but halves the gap (+0.008); pricing each state at the full 2.5 (`vy9t2f`) only returns the gap to 3b's (+0.015, nine of ten seeds) at lower room (19%). The reason is in the ceilings: with an independent second type a firm's total exposure is the sum of two ladder draws, six evenly spread values that size and momentum proxy almost linearly, so the linear ceiling rises more than the nonlinear one (0.346 to 0.387). Both states act through the same channel, the growth-option elasticity, so the second state adds premium and a factor but not a second interaction. BGN's nine-type version (`bvy9t2`, +0.012, room 16%) sits between 3b and the three-direction cell. The reduced form (Table 6) says the same with the structure controlled.

Reduced form: bounded and curvature loadings

The RF rows of Table 8 are the `kfh` ladder with bounded (tanh) and curvature (square) loadings on observed characteristics, and the curvature ladder with the loading directions orthogonal to size (`kfo`); the main table (Table 6) keeps only the linear loadings and the latent products.

Bounded loadings behave like linear ones: no room at $K = 1$ and 4, a little at $K = 16$ (12%), and a negative gap throughout, because tanh of a single index is monotone in that index and nearly linear in its rank. Pure curvature of an observed characteristic gives room of about five times the linear ceiling and a gap of about +0.5 at $K = 1$. In the unconstrained square rows the gap then falls to +0.12 and +0.14 at $K = 4$ and 16 while the room does not move: the squared index contains size^2, Fama–French's legs are value-weighted and market value is exp(size), so those legs hold a size^2 exposure mechanically and Fama–French's Sharpe jumps from 0.05 to 0.46. The `kfo` rows remove the size coordinate from every direction and the gap returns to about +0.5 at every K. Curvature of an observed characteristic is therefore captured by random features and by nothing linear, and the square ladder's apparent fall with K was an artefact of value weighting.